

20. **Ans D**

When switch is open,

$$V_{PQ} = (0.70/1.0) \times 2.0 \text{ V} = 1.4 \text{ V}$$



$$E = 1.4 \text{ V}$$

When switch is closed,

$$V_{PQ} = (0.50/1.0) \times 2.0 \text{ V} = 1.0 \text{ V}$$

By potential divider principle,

$$\text{p.d. across } 2.0 \, \Omega, V_{2.0} = (2.0/(2.0+r)) \times 1.4 \text{ V}$$

$$1.0 = (2.0/(2.0+r)) \times 1.4$$



$$r = 0.80 \, \Omega$$

21. **Ans: C**